Diseases of the Thyroid Gland

NURA 340 Pathophysiology | Final Exam Lecture Study Guide

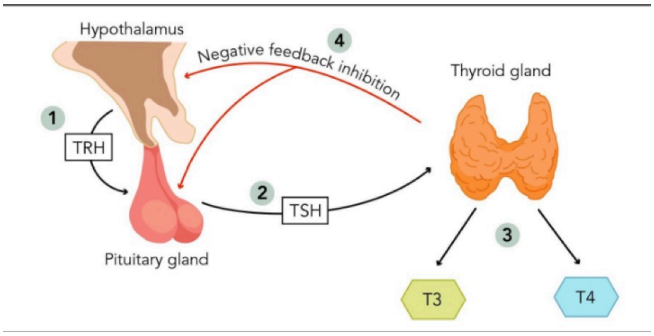
Hypothyroidism, hyperthyroidism, goiter, and thyroid nodules

Lecture roadmap: Normal thyroid regulation (the feedback loop) → Hypothyroidism → Primary vs. secondary vs. tertiary vs. subclinical → Hyperthyroidism (Graves', nodules, thyroiditis) → Atrial fibrillation risk → Why TSH moves opposite to T3/T4 → Goiter → Thyroid nodules → Why enlargement causes dysphagia and hoarseness

How the instructor wants you to think: Thyroid hormone controls how the body **uses energy**, so a change affects almost every system. If you know what thyroid hormone normally does, the findings explain themselves: **hypo = everything slows down**, **hyper = everything speeds up**. For labs, **always go back to the feedback loop**: ask where the problem starts and how a working pituitary *should* respond. Reason through the results instead of memorizing them.

How to read this guide: Everything without a tag comes from the lecture slides and transcript. Anything else is tagged: ADDED information not in the lecture; CLARIFIED a lecture point explained further or a transcript slip cleaned up; CORRECTED an error in the slides or transcript. Every tag is listed in the notes at the end.

Normal Thyroid Regulation



From the lecture slides: the hypothalamic–pituitary–thyroid axis.

1. **Hypothalamus** releases **TRH** (thyrotropin-releasing hormone).
2. TRH stimulates the **anterior pituitary** to release **TSH** (thyroid-stimulating hormone).
3. TSH stimulates the **thyroid** to make and release **mostly T4** and a **small amount of T3**. Many tissues convert T4 into **T3, the more active form**.

4. **Negative feedback:** when enough thyroid hormone is circulating, the hypothalamus and pituitary **reduce** their stimulation. When levels fall, stimulation **increases**.

T3 and T4 regulate **metabolic activity**: energy use, **heat production**, **cardiac function**, digestion, brain development, and more.

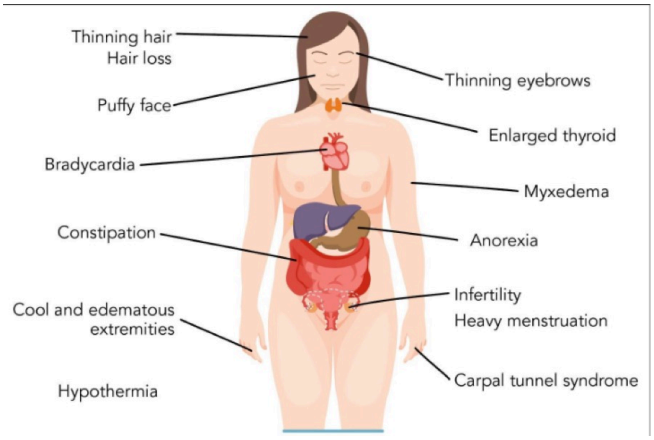
> **Keep track of where each hormone comes from (instructor emphasis): TSH comes from the pituitary. T3 and T4 come from the thyroid.** This is the key to reading thyroid labs.

Hypothyroidism

There isn't enough thyroid hormone to meet the body's needs, so **many body processes slow down**.

Etiology	**Primary** (problem in the thyroid itself; most common): • **Hashimoto's disease**: **autoimmune**. The immune system attacks and damages thyroid tissue, so over time the gland can't make enough hormone. • **Surgery**, certain **medications**, and **treatments that destroy thyroid tissue** • **Iodine deficiency**: the thyroid needs iodine to make its hormones CORRECTED **Central** (problem higher up in the loop, in the **pituitary or hypothalamus**): the thyroid could make hormone but **isn't being stimulated enough**.
Manifestations	Think "everything slows down": • **Fatigue**, lack of energy • **Cold intolerance**: the body produces less heat • **Constipation**: slower intestinal movement • **Bradycardia**: slower heart rate • **Dry skin, thinning hair** • **Weight gain** from reduced energy use and fluid retention • **Slowed thinking**: "foggy," trouble concentrating • **Heavier or irregular menstrual periods**; infertility These develop **gradually**, so a person may not realize the symptoms are connected. Look at the overall pattern.

Diagnosis	• **Overt primary hypothyroidism: ↑TSH with ↓free T4.** The thyroid isn't making enough, so the pituitary releases **more TSH** to push it. • **Free T4** = the portion of T4 **not attached to a binding protein**. It is available to enter tissues, which makes it useful for judging thyroid function. • **Thyroid antibodies** help identify an autoimmune cause (Hashimoto's).
Impact on health	Depends on how **severe and prolonged** the deficiency is: fatigue and slow thinking interfere with daily life; **elevated cholesterol**, **cardiovascular problems**, **reproductive difficulties**. **Myxedema coma**: a **life-threatening** state of severe hypothyroidism with **hypothermia**, **slowed breathing**, and **altered consciousness**. Despite the name, the person doesn't have to be in an actual coma. **Congenital hypothyroidism**: thyroid hormone is essential for **early brain development**, so **newborn screening** tests for it. Early treatment prevents developmental injury.



From the lecture slides: manifestations of hypothyroidism.

"Hypo" but high TSH: the classic confusion. In hypothyroidism you might expect everything to be low, but the lab used to screen for it is a **high TSH**. Why? The **pituitary is working normally** and turns TSH **up** to make a failing thyroid work harder. Think back to the negative feedback loop.

Primary, Secondary, Tertiary, and Subclinical (Reflection Question)

Work down the pathway: **hypothalamus** → **pituitary** → **thyroid**.

Type	Where the problem starts	Lab pattern	Clues
Primary	**Thyroid gland** (e.g., Hashimoto's)	↓ **free T4,** ↑ **TSH**	History of thyroid disease or thyroid treatment
Secondary	**Anterior pituitary**: not enough effective TSH	↓ **free T4 with low or "inappropriately normal" TSH**. These two labs alone usually **can't tell** secondary from tertiary.	Known pituitary or hypothalamic disorder, or **other hormone deficiencies** → points to a **central** cause
Tertiary	**Hypothalamus**: too little TRH → less pituitary stimulation → less thyroid stimulation		
Subclinical	**Not a location.** It's a **lab pattern**.	↑ **TSH, free T4 still normal**	Mild or **no** symptoms, so labs give the bigger hint

- **"Inappropriately normal" TSH:** if thyroid hormone is low, a normal TSH is the *wrong* response. A working pituitary should have raised it, so the signaling system is part of the problem.
- Fatigue, cold intolerance, constipation, and bradycardia reflect **low hormone**. They occur in all three types and **don't tell you where** the problem started.
- Put together the **symptoms, history, and labs**. Ask: Is thyroid hormone low? Is the pituitary **responding the way it should**?

Hyperthyroidism

The thyroid makes **more hormone than the body needs**, so **body processes speed up**.

> **Two terms:** **Hyperthyroidism** = the **thyroid itself is making too much**. **Thyrotoxicosis** = the **effects of excess thyroid hormone from any source**. The main question is **where the extra hormone is coming from**: more production, or release of stored hormone? That's why treatment depends on the cause.

Etiology and pathophysiology	• **Graves' disease**: **autoimmune**, but unlike Hashimoto's the antibodies **stimulate** the thyroid. They bind the **same receptors TSH normally binds** and tell the gland to keep making hormone. The pituitary lowers TSH, but the antibodies keep stimulating anyway, overriding the feedback loop. • **Thyroid nodules**: some function **independently** and make hormone without the TSH signal. • **Thyroiditis**: inflammation damages thyroid tissue and **already-stored hormone leaks** into the blood. Excess hormone effects occur even though production hasn't increased. When the stores run out, levels may fall, sometimes into a period of **hypothyroidism**.
Manifestations	Think "everything speeds up": • More energy use and **heat production** → feels **warm and sweaty** (heat intolerance) • **Weight loss** even when eating normally or feeling hungrier • **More frequent bowel movements or diarrhea** (increased intestinal activity) • **Restless, anxious, shaky**; **trouble sleeping** • **Muscle weakness**: a faster metabolism doesn't mean the person feels energetic or strong • **Palpitations, tachycardia**: thyroid hormone raises heart rate and makes the heart **more responsive to sympathetic stimulation** • **Graves' only: thyroid eye disease**. Autoimmune inflammation of the tissues around the eyes causes discomfort, swelling, or **protruding eyes**. It comes from the autoimmune process and is **not** a finding in all hyperthyroidism.
Diagnosis	**Overt primary hyperthyroidism: ↓TSH with ↑free T4 and/or T3.** The pituitary turns its signal down because there's already too much hormone. More testing identifies the cause.
Impact on health	• Increased **cardiac workload** and ↑ **risk of atrial fibrillation** (see below) • **Bone loss** from persistent excess • **Thyroid storm**: thyrotoxicosis causes **acute systemic decompensation** with **high fever**, **marked tachycardia**, and **mental status changes**. This is far more serious than usual hyperthyroid symptoms.

Reflection: Is a client with untreated hyperthyroidism at greater risk for atrial fibrillation? Why?

Yes. Excess thyroid hormone **raises heart rate**, makes the heart **more responsive to sympathetic stimulation**, and **changes electrical activity** in cardiac tissue, so an abnormal rhythm like AFib becomes more likely. In AFib, atrial electrical activity is **disorganized** and the atria lose their coordinated contraction → **less ventricular filling** and **blood pooling** in the atria → **clots and stroke**. Not everyone develops AFib, but this is why **palpitations and tachycardia deserve attention** in hyperthyroidism.

Reflection: Why do TSH levels move opposite to T3/T4 in primary thyroid disorders?

TSH comes from the **pituitary**; T3 and T4 come from the **thyroid**. In **negative feedback**, the pituitary adjusts TSH to the amount of circulating thyroid hormone:

- **Primary hypo:** too little hormone → pituitary **raises TSH** ("work harder") → **↑TSH, ↓free T4**
- **Primary hyper:** too much hormone → pituitary **lowers TSH** → **↓TSH, ↑free T4/T3**

The word **primary** matters: these patterns assume the problem is in the thyroid and the **pituitary responds normally**. If the problem is in the pituitary or hypothalamus, that opposite relationship may be lost. Ask: **does the TSH response make sense for the amount of thyroid hormone present?**

Hypo vs. Hyper at a Glance

Feature	Hypothyroidism (slow)	Hyperthyroidism (fast)
Autoimmune cause	**Hashimoto's**: antibodies **damage** the thyroid	**Graves'**: antibodies **stimulate** the thyroid
Temperature	Cold intolerance	Warm, sweaty
Weight	Gain	Loss (despite normal or increased appetite)
Bowels	Constipation	Frequent BMs, diarrhea
Heart	Bradycardia	Tachycardia, palpitations, AFib risk
Mind	Foggy, slow thinking	Restless, anxious, shaky, insomnia

Feature	Hypothyroidism (slow)	Hyperthyroidism (fast)
Other	Dry skin, thinning hair, heavy/irregular periods, ↑cholesterol	Muscle weakness, bone loss; eye disease (Graves')
Primary labs	**↑TSH, ↓free T4**	**↓TSH, ↑free T4/T3**
Emergency	**Myxedema coma**: hypothermia, slow breathing, altered consciousness	**Thyroid storm**: high fever, marked tachycardia, mental status change

Structural Disorders: Goiter and Nodules

So far the focus was **how much hormone** is available. Now the focus is the gland's **size and structure**. A **goiter** is an enlarged thyroid; a **nodule** is a distinct lump within it. Both can occur with **normal, low, or high** thyroid function. **An enlarged thyroid does not automatically mean too much hormone.** For each, ask two things: **How well is the gland functioning?** and **Is the change in structure affecting nearby tissues?**

Goiter

Etiology and pathophysiology	The term says the gland is bigger but **not why**. Mechanisms: • **Prolonged TSH stimulation**, e.g., **iodine deficiency**: the thyroid struggles to make hormone → pituitary raises TSH → ongoing stimulation makes thyroid tissue **grow**. • **Autoimmune**: **Graves'** (antibodies stimulate the gland) or **Hashimoto's** (inflammation and tissue changes). • A goiter may contain **multiple nodules**.
Manifestations	May have **no symptoms** and be found on exam. Others notice **swelling at the front of the neck** or **pressure or tightness**. Because the thyroid sits next to the **trachea and esophagus**, enlargement can interfere with **breathing or swallowing**. **Voice changes** occur if nearby nerves are affected.
Diagnosis	Evaluate both **function and structure**: thyroid **blood tests** (normal, increased, or decreased production); **antibody testing** (autoimmune cause); **ultrasound** (size, structure, nodules).

Impact on health	Depends on the cause and degree of enlargement: symptoms of **abnormal hormone levels**, symptoms of **compression**, or both. **Don't automatically assign a hormone pattern to a goiter.**

Thyroid Nodules

Etiology	A **localized growth** in the gland. There may be one or several, and they can be **solid, fluid-filled, or mixed**. More common with **age**; **most are benign**. More concern with a history of **radiation to the thyroid, especially in childhood**, or a **family history of thyroid cancer**.
Manifestations	Many cause **no symptoms** and are found on a routine exam or on imaging done for another reason (an **incidental finding**). Some make hormone **independently** → **hyperthyroidism**. Many have little effect on hormones. Large nodules → **visible lump**, **neck pressure**, **difficulty swallowing**. **Persistent hoarseness** raises concern that nearby structures are involved.
Diagnosis	Two questions: **Is it producing excess hormone? Are there features concerning for malignancy? TSH** assesses function; **ultrasound** evaluates size and characteristics; if **TSH is low**, additional scans show whether the nodule is actively making hormone.
Impact on health	Possible **hormone excess** or **compression**, though many cause neither. Waiting to learn whether a growth is benign causes **significant emotional stress**.

Reflection: Why might a goiter or nodule cause dysphagia, hoarseness, or throat fullness?

Picture the anatomy: the thyroid sits **just below the larynx**, with its lobes **alongside the trachea** and the **esophagus right behind**. A large enough goiter or nodule **presses on these structures**:

- Pressure on the **esophagus** → **dysphagia**
- The enlargement itself → **fullness or tightness** in the throat
- Pressure on **nerves controlling the vocal cords** → **hoarseness** CLARIFIED
- Pressure on the **trachea** → **breathing difficulty**

These are **physical effects of size and location** and can happen even when **hormone levels are normal**. Always consider both hormone function and compression.

Instructor wrap-up: Hyperthyroidism → think **increased metabolic activity and effects on the heart**. Hypothyroidism → think **insufficient hormone slowing body processes**. Goiters and nodules → consider **hormone function and the physical effects of enlargement**. Always return to the **feedback loop** for labs: where does the problem begin, and how should a functioning pituitary respond?

Self-Check ADDED

1. Which hormone comes from the pituitary, and which from the thyroid?
   TSH comes from the anterior pituitary. T3 and T4 come from the thyroid.
2. Which thyroid hormone is more active, and where is most of it made?
   T3. The thyroid releases mostly T4, which tissues convert to T3.
3. A client has TSH 12 (high) and low free T4. What is the most likely problem?
   Primary hypothyroidism. The thyroid is failing and the pituitary is compensating.
4. A client has low free T4 and a normal TSH. Why is that "normal" TSH abnormal?
   With low thyroid hormone, a working pituitary should raise TSH. Normal TSH is inappropriate and suggests a central (pituitary or hypothalamic) cause.
5. What is subclinical hypothyroidism?
   ↑TSH with normal free T4; mild or no symptoms. It is a lab pattern, not a location.
6. Hashimoto's vs. Graves': how do the antibodies differ?
   Hashimoto's antibodies damage the thyroid (hypo). Graves' antibodies bind the TSH receptor and stimulate it (hyper).
7. How can thyroiditis cause hyperthyroid symptoms without increased production?
   Inflamed tissue leaks stored hormone into the blood. Later the stores run out, which can cause hypothyroidism.
8. Name one finding seen only in Graves', not in all hyperthyroidism.
   Thyroid eye disease (protruding, swollen eyes).
9. Name the life-threatening emergency of severe hypothyroidism and of hyperthyroidism.
   Myxedema coma (hypothermia, slow breathing, altered LOC) and thyroid storm (high fever, marked tachycardia, mental status changes).
10. How does iodine deficiency cause a goiter?
    Low hormone → ↑TSH → prolonged stimulation makes thyroid tissue grow.
11. A client with a large goiter has normal thyroid labs but trouble swallowing. Is that possible?
    Yes. Compression of the esophagus is a physical effect of size, independent of hormone levels.

Notes on Sources, Additions & Clarifications

This guide was written from the lecture slides (Diseases of the Thyroid Gland, NURA 340) and the lecture transcript. The slides contain mainly headings and images, so most content comes from the narration.

- CORRECTED **Iodine deficiency (hypothyroidism causes).** The transcript says "iron deficiency because the thyroid needs iodine to make its hormones." The reasoning shows the lecturer meant **iodine** deficiency, which is how the guide lists it. The lecture uses iodine deficiency again correctly in the goiter section.
- CLARIFIED **Nerves and hoarseness.** The lecturer says enlargement can affect "nearby nerves that control vocal cord movement." These are the **recurrent laryngeal nerves**; the name is included for reference only.
- CLARIFIED **Weight gain wording.** The lecturer corrects "reduced energy" to "reduced **energy use**." The guide uses the corrected wording.
- CLARIFIED **Heat intolerance** is the standard name for the "unusually warm and sweaty" finding in hyperthyroidism.
- ADDED **Hypo vs. Hyper at a Glance** table: it reorganizes lecture content side by side; the content itself is from the lecture.
- ADDED **Self-Check questions** are my own, written from lecture content.
- Calcitonin and the parathyroid glands appear in slide images but aren't discussed in the narration, so they are not covered here.